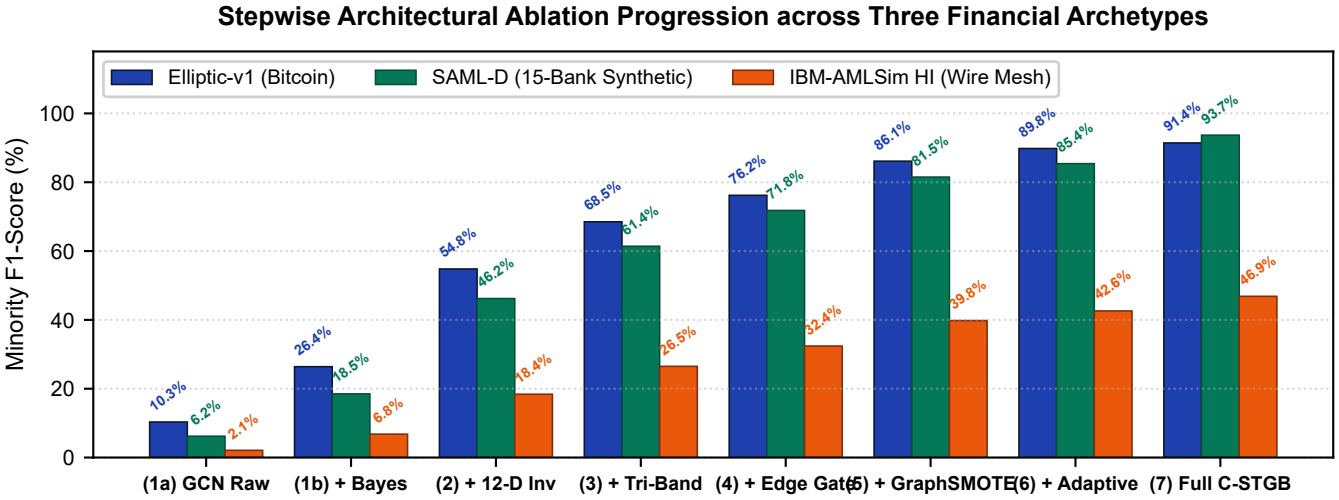
Stepwise Architectural Ablation Progression across Three Financial Archetypes
Elliptic-v1 (Bitcoin)
SAML-D (15-Bank Synthetic)
IBM-AMLSim HI (Wire Mesh)
Minority F1-Score (%)
0
20
40
60
80
100
10.3%
6.2%
2.1%
26.4%
18.5%
6.8%
54.8%
46.2%
18.4%
68.5%
61.4%
26.5%
76.2%
71.8%
32.4%
86.1%
81.5%
39.8%
89.8%
85.4%
42.6%
91.4%
93.7%
46.9%
(1a) GCN Raw
(1b) + Bayes
(2) + 12-D Inv
(3) + Tri-Band
(4) + Edge Gat
(5) + GraphSMOTE
(6) + Adaptive
(7) Full C-STGB